CSE STUDY HUB

Communication Skills for Engineers II — Aptitude

UNIT 4 · Financial Aptitude

What this unit is about

This unit builds understanding of **Financial Aptitude**. These notes are designed for quick learning and exam revision: learn the meaning, understand the working, practise an example, and finish with comparisons and key terms.

Syllabus coverage checklist

1. Cost price, selling price, profit and loss

2. Profit/loss percentage calculated on cost price

3. Discount, marked price and successive discounts

4. Faulty weights and dishonest transactions

5. Simple interest: SI = PRT/100; amount = P + SI

6. Finding principal, rate or time from simple-interest data

7. Compound interest: A = P(1 + R/100)^T for annual compounding

8. Difference between SI and CI; compounding frequency

How to study: Read each topic once for meaning, underline definitions and formulas, reproduce diagrams/steps without looking, then answer the revision questions on Page 5.

Exam answer pattern

For a 5-mark answer: (1) definition, (2) labelled diagram or formula, (3) working/components in steps, (4) example/application, and (5) one advantage, limitation or comparison.

UNIT 4 NOTES · PART 1

Cost price, selling price, profit and loss

Cost price, selling price, profit and loss is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Profit/loss percentage calculated on cost price

x percent means x per 100. To increase by r%, multiply by (1+r/100); to decrease, multiply by (1-r/100). For successive changes multiply the factors rather than adding the percentages.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Discount, marked price and successive discounts

Discount, marked price and successive discounts is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Financial Aptitude, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 2

Faulty weights and dishonest transactions

Faulty weights and dishonest transactions is a syllabus concept in Communication Skills for Engineers II — Aptitude. Learn its definition, purpose, key terms and working principle. Organise the explanation into components or steps, then add a practical example, one advantage, one limitation and a comparison with a related concept.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Simple interest: SI = PRT/100; amount = P + SI

Simple interest is calculated only on the original principal: SI = P×R×T/100 when R is annual and T is in years. Amount = P+SI. Keep rate and time units consistent.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Finding principal, rate or time from simple-interest data

For Finding principal, rate or time from simple-interest data, identify the given values and the quantity to find before selecting a formula. Keep units consistent, show substitutions, calculate carefully and perform a reasonableness check. Write the formula and one solved example in your revision copy.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Financial Aptitude, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

UNIT 4 NOTES · PART 3

Compound interest: A = P(1 + R/100)^T for annual compounding

For annual compounding, A=P(1+R/100)^T and CI=A–P. Interest is added to principal each period, so later interest is earned on earlier interest too.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Difference between SI and CI; compounding frequency

Compare the alternatives in Difference between SI and CI; compounding frequency using the same criteria: purpose, working method, performance, cost/complexity, security or reliability, and suitable use case. End with a short conclusion stating when each option is preferable.

Write in your own words: Define the concept, list its important parts/steps, and attach one example from a system, program, workplace or numerical problem.

Connect and apply

For a practical question on Financial Aptitude, first identify which syllabus concept is being tested. State assumptions, draw the relevant structure/flow, work through the steps in order, and finish by interpreting the result. Mention complexity, reliability, accuracy, cost or ethical impact whenever it is relevant.

QUICK REVISION & EXAM PRACTICE

Key takeaways

• **Cost price, selling price, profit and loss** — know the definition, core working and one example.

• **Profit/loss percentage calculated on cost price** — know the definition, core working and one example.

• **Discount, marked price and successive discounts** — know the definition, core working and one example.

• **Faulty weights and dishonest transactions** — know the definition, core working and one example.

• **Simple interest: SI = PRT/100; amount = P + SI** — know the definition, core working and one example.

• **Finding principal, rate or time from simple-interest data** — know the definition, core working and one example.

• **Compound interest: A = P(1 + R/100)^T for annual compounding** — know the definition, core working and one example.

• **Difference between SI and CI; compounding frequency** — know the definition, core working and one example.

Worked-answer framework

Scenario: A question asks you to explain or apply a concept from Financial Aptitude. **Step 1:** identify the relevant term and define it. **Step 2:** write the formula, components or ordered process. **Step 3:** apply it to the given data/scenario. **Step 4:** state the result and justify it. **Step 5:** mention one limitation or precaution.

Self-test questions

Q1. Define Cost price, selling price, profit and loss and explain its importance.

Q2. Explain the working/steps involved in Profit/loss percentage calculated on cost price.

Q3. Differentiate between two related concepts from Financial Aptitude.

Q4. Give one practical application of Difference between SI and CI; compounding frequency and discuss one limitation.

Q5. Write a structured 5-mark answer covering the main ideas of Financial Aptitude.

Final checklist: I can define every topic · explain the process without notes · reproduce formulas/diagrams · solve one example · compare related concepts.